

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Bachelor of Technology (Civil Engineering) SEMESTER - 4 Winter 2025 (Supply.)

Course :Bachelor of Technology (Civil Engineering) Branch : Engineering and Technology

Semester : SEMESTER - 4

Subject Code & Name: BTCVC405 - HYDRAULICS - II

Time : 3 Hours]

[Total Marks : 60

Instructions to the Students:

1. Each question carries 12 marks.
 2. Question No. 1 will be compulsory and include objective-type questions.
 3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6
 4. Use of non-programmable scientific calculators is allowed.
 5. Assume suitable data wherever necessary and mention it clearly.
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- Q1. Objective type questions. (Compulsory Question) 12
- 1 Which weir is most suitable for precise discharge measurement?
 - a) Broad crested
 - b) Ogee
 - c) Sharp crested
 - d) Submerged
 - 2 Velocity of flow in open channel is commonly measured using:
 - a) Venturimeter
 - b) Pitot tube
 - c) Current meter
 - d) Orifice meter
 - 3 A most economical channel section has:
 - a) Maximum wetted perimeter
 - b) Minimum wetted perimeter
 - c) Maximum hydraulic gradient
 - d) Minimum velocity
 - 4 Critical flow in open channel occurs when Froude number is:
 - a) < 1
 - b) > 1
 - c) $= 0$
 - d) $= 1$
 - 5 Hydraulic jump is an example of:
 - a) Uniform flow
 - b) Gradually varied flow
 - c) Rapidly varied flow
 - d) Laminar flow
 - 6 Impact of jet on vanes is based on:
 - a) Energy equation
 - b) Continuity equation
 - c) Momentum principle
 - d) Bernoulli's theorem

- 7 Turbine suitable for low head and high discharge is:
 - a) Pelton
 - b) Francis
 - c) Kaplan
 - d) Turgo
- 8 Draft tube is used in reaction turbines to:
 - a) Increase velocity
 - b) Recover kinetic energy
 - c) Reduce discharge
 - d) Increase friction loss
- 9 Which pump is preferred for deep bore wells?
 - a) Centrifugal
 - b) Reciprocating
 - c) Submersible
 - d) Gear pump
- 10 Chezy's equation is used to compute:
 - a) Discharge
 - b) Velocity of flow
 - c) Hydraulic radius
 - d) Channel slope
- 11 Manning's roughness coefficient depends mainly on:
 - a) Channel slope
 - b) Flow velocity
 - c) Channel material
 - d) Discharge
- 12 GVF profiles depend on:
 - a) Reynolds number
 - b) Channel slope and depth
 - c) Discharge only
 - d) Channel width

Q2. Solve the following.

- A) Explain energy equation as applied to open channel flow. 6
- B) Write the differences between pipe flow and open channel flow? 6

Q3. Solve the following.

- A) Derive the condition for most economical rectangular channel section 6
- B) A rectangular channel carries a discharge of $20 \text{ m}^3/\text{s}$. Determine the critical depth of flow. 6

Q4. Solve Any Two of the following.

- A) Define gradually varied flow. Explain assumptions made in GVF analysis. 6
- B) Explain the phenomenon of hydraulic jump with neat sketch. State its applications. 6
- C) A jet of water 60 mm in diameter issues with a velocity of 20 m/s and strikes normally on a smooth flat plate. 6
 Determine the force exerted by the jet when the plate is:
 - i) Stationary
 - ii) Moving with a velocity of 6 m/s in the direction of the jet
 Assume the density of water as 1000 kg/m^3 .

Q5. Solve Any Two of the following.

- A) Explain with neat sketch the construction and working of the Francis turbine. 6
- B) Distinguish between Pelton turbine and Kaplan turbine. 6
- C) A Pelton wheel operates under a head of 60 m and discharge of $0.8 \text{ m}^3/\text{s}$. Find the power developed if jet velocity coefficient is 0.98. 6

Q6. Solve Any Two of the following.

- A) Explain construction and working of a centrifugal pump with neat sketch. 6
- B) State advantages of centrifugal pump over reciprocating pump. 6
- C) A centrifugal pump is designed to operate against a net head of 18 m at a speed of 1200 rpm. The pump has backward curved vanes making an angle of 25° with the tangent at outlet. The impeller diameter at outlet is 350 mm and the width at outlet is 60 mm. If the manometric efficiency of the pump is 90%, determine the discharge of the pump. 6

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